

Musa Jawad

825-431-4976 | musajawad72@gmail.com | [linkedin.com/in/musa-jawad-208807239](https://www.linkedin.com/in/musa-jawad-208807239)

SUMMARY

Backend-focused Software Engineer with hands-on experience building production systems used by 1,500+ users. Specialized in API design, distributed systems, and performance optimization. Shipped real-world features in Agile teams using JavaScript, Python, SQL, and cloud platforms.

EDUCATION

University of Calgary

Bachelor of Science in Software Engineering

Canada, Calgary, AB

2025

EXPERIENCE

Fullstack Developer Intern

December 2024 – May 2025

Self Justice

Remote

- Worked in an Agile Scrum team to design and deliver a legal information platform serving 1,500+ active users.
- Shipped production backend services used by 1,500+ users, designing RESTful APIs in JavaScript that reduced manual legal deadline tracking by 90%
- Identified and eliminated backend bottlenecks, cutting form processing time by 50% and reducing user drop-off
- Increased automated test coverage from 60% to 90% through code reviews and unit testing.

Backend Developer Intern

June 2024 – December 2024

Inwit

Remote

- Built backend-driven web pages in PHP using the MVC design pattern, supporting 5,000+ monthly users with zero downtime.
- Optimized SQL queries and data access logic, reducing page load times by 35%.
- Automated server-side scripts and error handling, decreasing manual intervention by 80%.
- Delivered features ahead of schedule within Agile sprints, improving delivery efficiency by 25%.

PROJECTS

MediScanner | React, Node.js, Firebase, PaddleOCR, Capacitor, Axios

September 2023 – April 2025

- Built an accessibility-focused web and mobile application to extract Drug Identification Numbers (DINs) from medication images, achieving 95.7% accuracy across 500+ test cases.
- Integrated external APIs including the Canadian Drug Product Database and Perplexity AI for real-time medication lookup, reducing manual errors by 70%.
- Designed an image-processing pipeline using OCR preprocessing and regex-based validation, improving OCR throughput by 40%.
- Implemented authentication, storage, and deployment using Firebase to support secure, cross-platform access.

Distributed Agar.io Clone | Phaser, TypeScript, Django, SQL, WebSockets, HTTP

January 2025 – April 2025

- Designed and implemented a distributed real-time multiplayer system supporting 75+ concurrent users with sub-100ms latency using WebSockets and replicated SQL backends.
- Architected a layered backend system with a proxy server, Django-based primary replica, Lamport timestamps, and multiple synchronized SQL replicas.
- Implemented real-time communication using WebSockets to synchronize player state, improving responsiveness by 60% compared to polling.

TECHNICAL SKILLS

- **Languages:** Java, JavaScript, Python, C/C++, SQL (PostgreSQL), PHP, HTML/CSS, R
- **Frameworks & Runtimes:** React, Node.js, Django, Flask, FastAPI, PHPUnit, JUnit, Material-UI
- **Architecture & Methodologies:** RESTful APIs, MVC, Layered Architecture, Distributed Systems, Agile (Scrum)
- **Developer Tools & Platforms:** Git, Docker, Google Cloud Platform (GCP), Firebase, VS Code, IntelliJ IDEA, PyCharm
- **Libraries:** Pandas, NumPy, Matplotlib